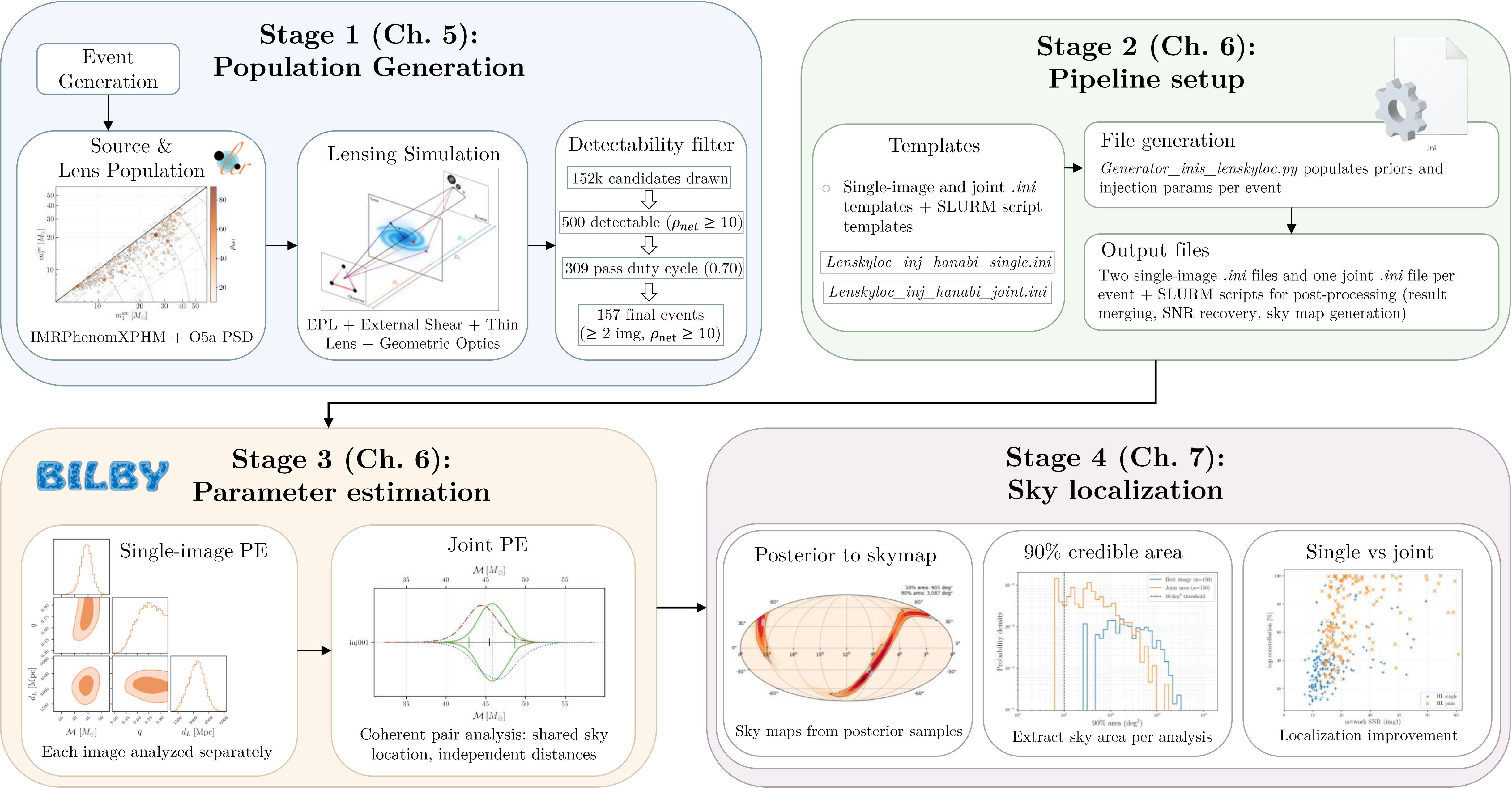

Stage 1 (Ch. 5):
Population Generation
Event Generation
Source & Lens Population
IMRPhenomXPHM + O5a PSD
Lensing Simulation
EPL + External Shear + Thin Lens + Geometric Optics
Detectability filter
152k candidates drawn
500 detectable ($\rho_{net} \geq 10$)
309 pass duty cycle (0.70)
157 final events ($\geq$ 2 img, $\rho_{net} \geq 10$)
Stage 2 (Ch. 6):
Pipeline setup
Templates
Single-image and joint .ini templates + SLURM script templates
Lenskyloc_inj_hanabi_single.ini
Lenskyloc_inj_hanabi_joint.ini
File generation
Generator_inis_lenskyloc.py populates priors and injection params per event
.ini
Output files
Two single-image .ini files and one joint .ini file per event + SLURM scripts for post-processing (result merging, SNR recovery, sky map generation)
BILBY
Stage 3 (Ch. 6):
Parameter estimation
Single-image PE
Each image analyzed separately
Joint PE
inj001
Coherent pair analysis: shared sky location, independent distances
Stage 4 (Ch. 7):
Sky localization
Posterior to skymap
50% area: 905 deg²
90% area: 3,087 deg²
Sky maps from posterior samples
90% credible area
Extract sky area per analysis
Single vs joint
Localization improvement